KPIT APEX LAB

Summer Internship Program | Academic Year 2025-2026

Safety Supervisor ECU

with Emergency Brake Assist

A prototype Automotive Safety ECU built on STM32, integrating LIDAR and IMU sensing with CAN-based emergency brake control.

Submitted By: Uzair Shaikh, Om Raut, Sahil Deshmukh

Guided By: Isheet Patel

Why an Emergency Brake Assist ECU?

Delayed driver reaction is one of the leading causes of rear-end collisions. Even an attentive driver needs time to perceive an obstacle, decide, and brake — during which the vehicle keeps moving. Advanced Driver Assistance Systems (ADAS), particularly Automatic Emergency Braking (AEB), close this gap by monitoring the surroundings and intervening automatically when required.

This project implements a prototype Safety Supervisor ECU on an STM32 Nucleo-F446RE. It continuously reads LIDAR and IMU sensor data, classifies the vehicle into a safety state, and issues emergency braking commands over CAN to a FireBird V robot acting as the experimental vehicle.

Core Idea

Sense distance → Classify risk → Alert the driver → Brake automatically, reproducing the core logic of modern AEB systems on an embedded, low-cost platform.

GOALS

Project Objectives

Real-Time Obstacle Sensing

Continuously measure distance to obstacles using the LIDAR Lite v3HP and monitor vehicle motion via the MPU6050 IMU.

Three-Level Safety Classification

Classify the vehicle into Normal, Warning, or Emergency states based on live sensor data.

CAN-Based Brake Control

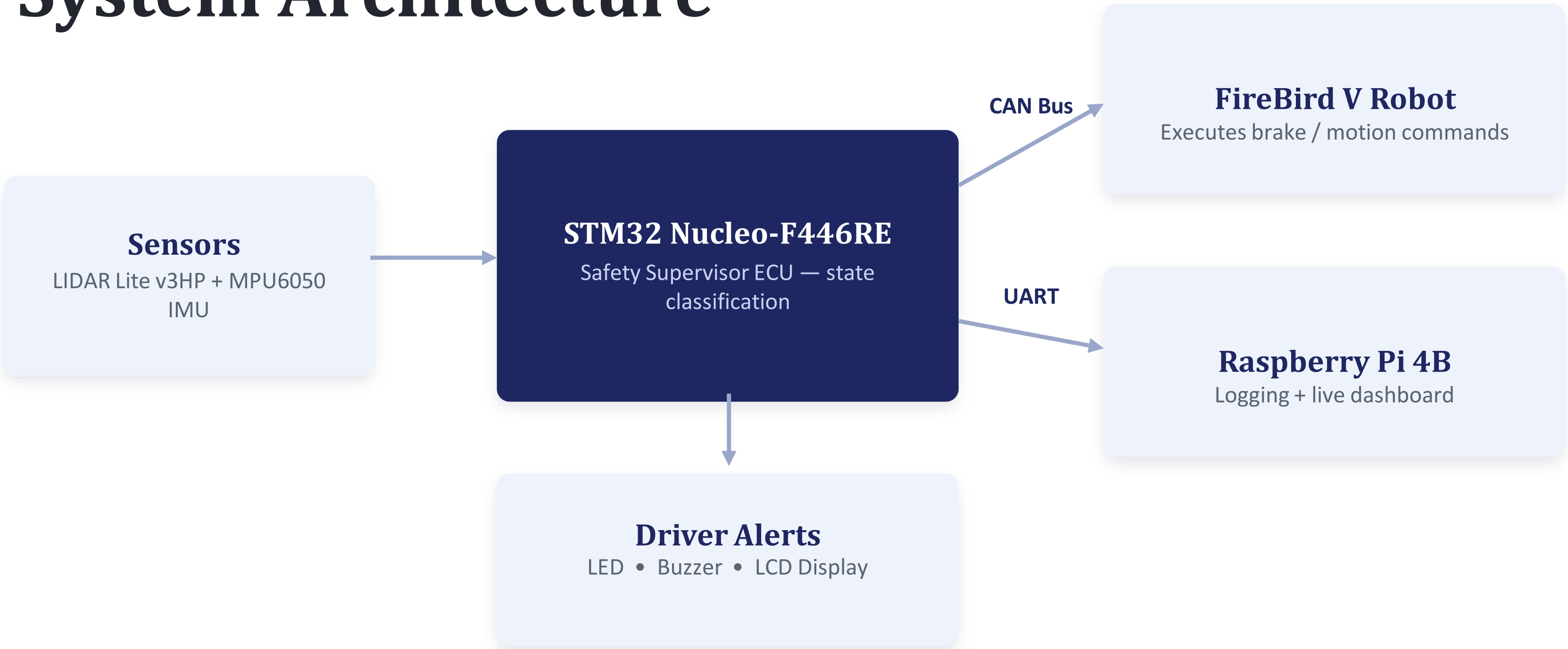
Transmit emergency braking commands from the STM32 ECU to the FireBird V robot over a CAN bus.

Logging & Visualization

Log sensor data and safety states on a Raspberry Pi and present them on a live dashboard.

DESIGN

System Architecture



Hardware Components

Component	Role in the System
STM32 Nucleo-F446RE	Central ECU — sensor processing, state logic, alerts & CAN control
FireBird V Robot	Experimental vehicle platform; executes braking and motion commands
LIDAR Lite v3HP	Laser sensor for real-time obstacle distance measurement (I2C)
MPU6050 IMU	6-axis accel + gyro for vehicle motion and deceleration data (I2C)
MCP2515 CAN Module (x2)	SPI-to-CAN controllers linking the STM32 ECU and FireBird V robot
Raspberry Pi 4B	Data logging, CSV storage and dashboard visualization (Python)
LCD, LEDs & Buzzer	Visual and audible driver-alert indicators for each safety state

BUILD

Software Technologies

Tool	Purpose
Arduino IDE/STM32CubeIDE	Firmware development, debugging and flashing for the STM32 ECU
STM32CubeMX	Graphical config of I2C, SPI, CAN, GPIO and timer peripherals
Atmel Studio	Programming and debugging the FireBird V robot's ATmega2560
Python	Raspberry Pi logging scripts, serial parsing and dashboard visuals
Raspberry Pi OS	Linux environment for logging and monitoring application

CORE LOGIC

Safety State Classification

NORMAL

Safe distance from obstacle

Full-speed operation continues; system keeps monitoring

WARNING

Obstacle getting closer

Yellow LED + buzzer alert; gradual motor speed reduction

EMERGENCY

Collision risk very high

Red LED + buzzer; controlled rapid motor stop via CAN

Hysteresis and moving-average filtering on the LIDAR readings prevent rapid oscillation between states near threshold boundaries.

CONTROL

CAN Communication & Brake Assist

Two MCP2515 CAN controllers — one on the STM32 ECU, one on the FireBird V — form a CAN bus between the two platforms. Brake commands are given the highest CAN message priority for fastest transmission, and bus-termination resistors keep signal integrity high.

Safety State	Command Sent Over CAN
Normal	Continue at current speed
Warning	Gradual reduction to ~50% speed
Emergency	Controlled rapid motor stop

Why CAN?

CAN is the standard, noise-resistant, multi-node protocol used across modern automotive networks — ideal for reliable safety-critical commands.

MONITORING

Data Logging & Dashboard

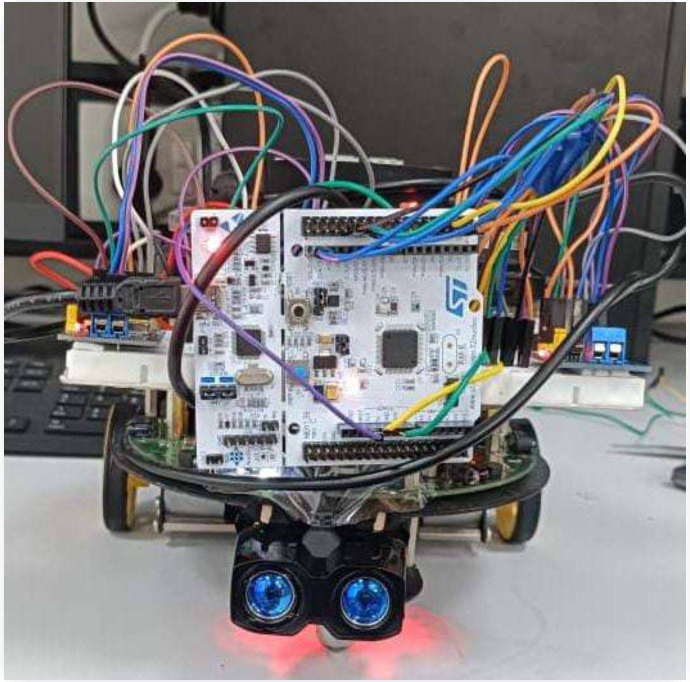
The STM32 ECU streams safety-state, distance and IMU data to a Raspberry Pi 4B over UART. A Python script parses and logs each reading to CSV with a timestamp, while a live dashboard visualizes vehicle state, PWM speed, brake status and a rolling event log.

- Structured UART packet with start/end markers
- Buffered CSV logging with timestamps
- Live dashboard: state, speed, brake status, events

Raspberry Pi dashboard — live safety state, PWM speed and event log

VALIDATION

Testing & Results



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Safety states validated
(Normal / Warning / Emergency)

2

CAN nodes communicating
(STM32 ECU ↔ FireBird V)

End-to-End

Pipeline verified — sensing to braking to logging

Extended

Continuous-run reliability testing completed

Validated through static and dynamic obstacle tests, with objects at varying distance and approach speed confirming correct state transitions and brake responses.

RELEVANCE

Real-World Application Scenarios

Scenario	Risk	ECU Response
Lead vehicle brakes suddenly	High	Emergency stop if time-to-collision is unsafe
Pedestrian crosses ahead	High	Immediate emergency braking; event logged
Child runs out near school zone	Very High	Highest priority; immediate stop + alert
Animal appears on road	High	Warning or Emergency based on distance/speed
Lead vehicle slows gradually	Medium	Warning state; alert driver, prepare to slow
Rain, fog or dust noise	Medium	Extra sampling; degraded safe mode if uncertain

WHAT'S NEXT

Future Scope

Multi-Sensor Fusion

Add thermal camera and radar with LIDAR for stronger obstacle classification

GPS Tracking

Route monitoring and fleet-level supervision

Cloud Logging

Remote, real-time access to logs via Wi-Fi / cellular

Drowsiness Detection

Camera-based driver monitoring as an added safety layer

Embedded AI

On-ECU object detection and predictive collision avoidance

FreeRTOS Integration

Deterministic task scheduling for safety-critical timing

SUMMARY

Conclusion

The Safety Supervisor ECU project successfully demonstrates a working prototype automotive safety system — combining real-time LIDAR/IMU sensing, three-level safety state classification, CAN-based emergency brake control, and Raspberry Pi data logging on a low-cost embedded platform.

- Reliable sensor integration and real-time decision-making
- Verified CAN-based communication between STM32 ECU and FireBird V robot
- Automatic emergency braking with multi-modal driver alerts
- Continuous data logging and live dashboard visualization

Thank You

Safety Supervisor ECU with Emergency Brake Assist

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